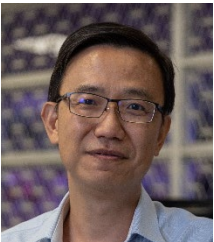




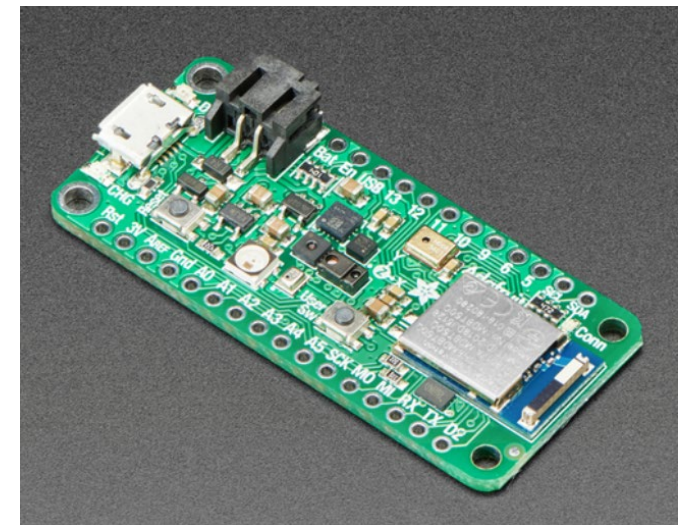
SC4172

Internet of Things - TinyML

Adafruit Feather Sense Board

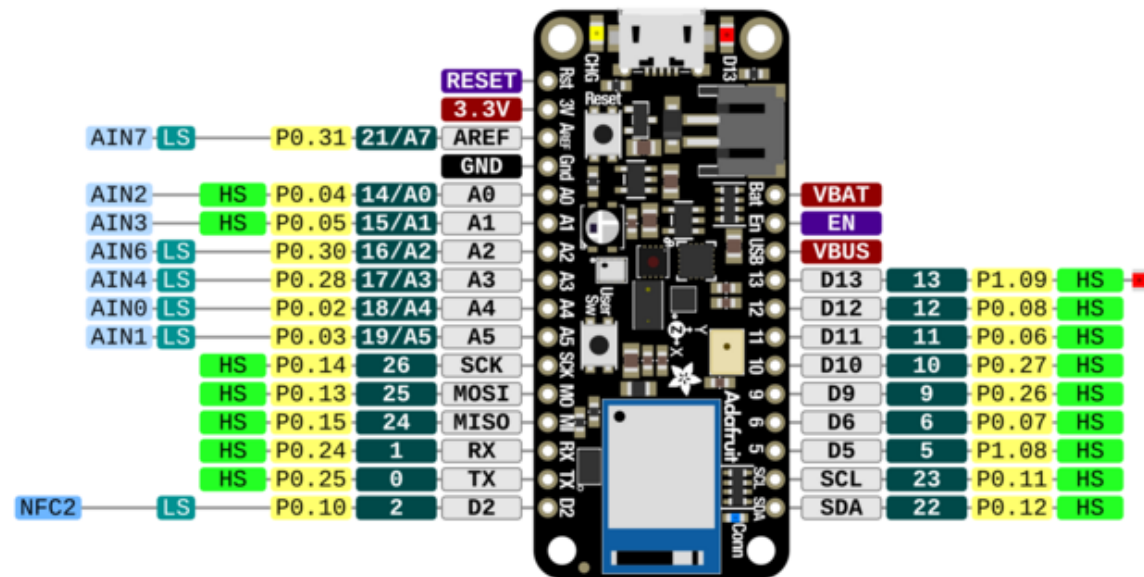


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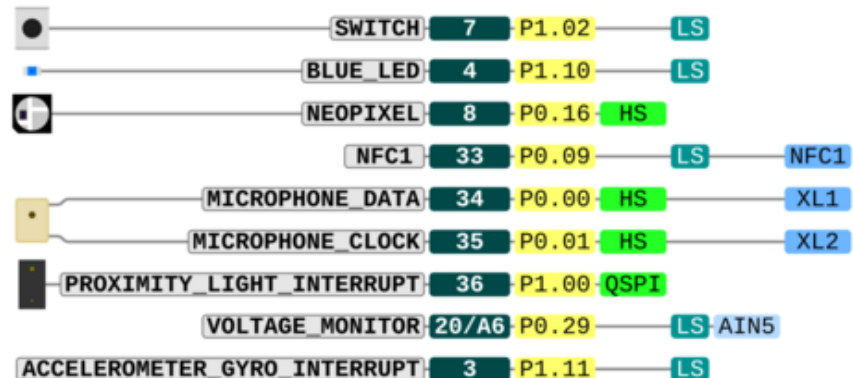


Adafruit Feather Sense Board

- Power
- GND
- Control
- CircuitPython Name
- Arduino Name
- GPIO
- High Speed
- Low Speed
- ADC
- Special



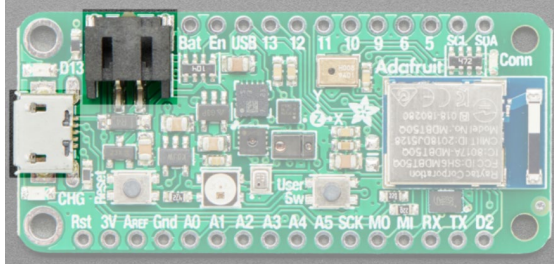
The Nordic nRF52840 is an ARM Cortex-M4 running at 64MHz with 256kB on-chip SRAM, 1MB of Flash memory and built-in Bluetooth Low Energy and native USB. Any pin can be used for I2C/SPI/UART/I2S/PDM/etc peripheral usage. LS pins can interfere with radio reception when over 10 KHz, however they are not restricted from going faster. All GPIO is 3.3V in/out max unless otherwise stated.



Adafruit Feather Sense Board

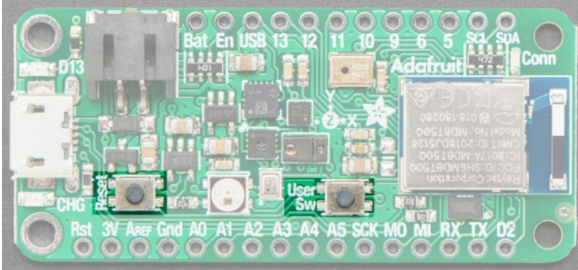
- Nordic Semiconductor System-On-Chip **nRF52840**
 - 32bit Cortex M4F CPU @64Mhz
 - Internal memory: 256KByte SRAM, 1MByte Flash
 - 3.3V I/O => **not 5V tolerant.**
- Sensors On board
 - 6DOF Gyro+Accelerometer: **LSM6DS33** or LSM6DS3TRC
 - Magnetometer: **LIS3MDL**
 - Light + Gesture + Proximity: **APDS9960**
 - Humidity: **SHT31**
 - Temp + Pressure: **BMP280**
 - PDM Microphone sound sensor: **MP34DT01-M**

Lipo Battery



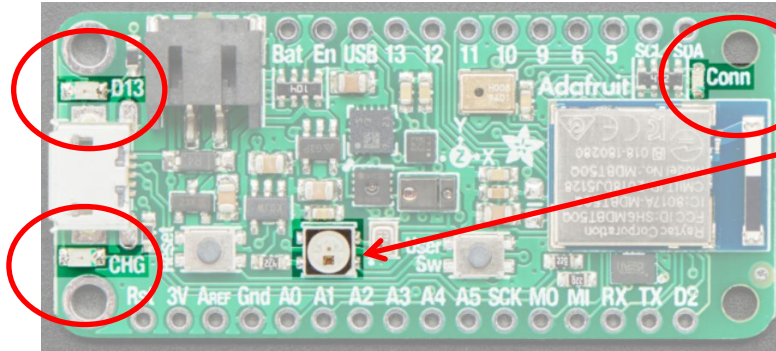
- Adafruit board can be power directly with Lipo battery connected via JST-PH 2-pin header.
- Battery can be **charged via USB**,
- Charging of battery and board operation can be done simultaneously.
- **Lipo battery is susceptible to fire hazard** when physically damaged, please handle with care. Do not leave the battery to charge without monitoring the process in-person.
- Student using this board need to assemble the **proto-board and spacers** to protect the Lipo battery. See above for reference.

Buttons



- **Reset button** - The button on the left next to the USB connector resets the board. Press **once to reset**. Quickly **press twice to enter the bootloader (DFU)**.
- **User button** - The button on the right is both usable by the bootloader and usercontrollable. Address it in code using **D7** in Arduino, i.e. **pin 7**.

LEDs



NeoPixel (sample codes available in Arduino sketch examples)

- **Red** status LED - This little red LED, labelled **D13**, works as a status LED in the bootloader. Otherwise, it is controllable using code by addressing D13 in Arduino.
- **Blue** status LED - This little blue LED, labelled Conn, works as a connectivity status LED in Arduino. Controllable using code by addressing **D4** in Arduino.
- **Amber** status LED - The little LED, labelled CHG, below the USB connector is the charge status LED. When no battery is connected, it flashes. When a battery is connected and charging, the LED is steady amber.

Arduino Sketch

- Board setup procedure outlined online in the following URL. Board package “**Adafruit nRF52**”
- <https://learn.adafruit.com/adafruit-feather-sense/arduino-support-setup>
- Libraries required for specific sensors can typically be located by searching for **Adafruit+sensor**_name in the library manger in Arduino sketch.
- Sample codes for each of the sensors are available in the library.
Files->Examples->SensorName
- Remember to select the correct COM Port before uploading the file to the hardware.
- Adafruit board requires it **bootloader** to be **re-flashed after user upload their code**. Note that Adafruit uses a different COM port to upload user code and its bootloader code.
- Arduino Library files is located under **Documents\Arduino\libraries**

CASE STUDY#1: BLE+SENSOR INTEGRATION

Adafruit Sensors Integration

- Sensorintegration Arduino Sketch can be downloaded from NTU Learn. The application reads from all the sensors on feather sense board, data can be output to Serial Monitor or transmitted to a BLE Central device.
- A list of defines allow individual sensors to be enabled/disabled.
- In addition, sample codes for individual sensors on the Feather Sense Board can be found in <https://learn.adafruit.com/adafruit-feather-sense/arduino-sensor-example>

```
//Features Enable/Disable
#define PROXIMITY 1
#define TEMP_BARO 1
#define MAGNETO 1
#define ACC_GYRO 1
#define HUMIDITY 1
#define MIC 0 //Do not use. Issue with programming DFU if this is enabled.
#define PRINT_SENSOR_OUTPUT 0
```

Links to detail information on sensors

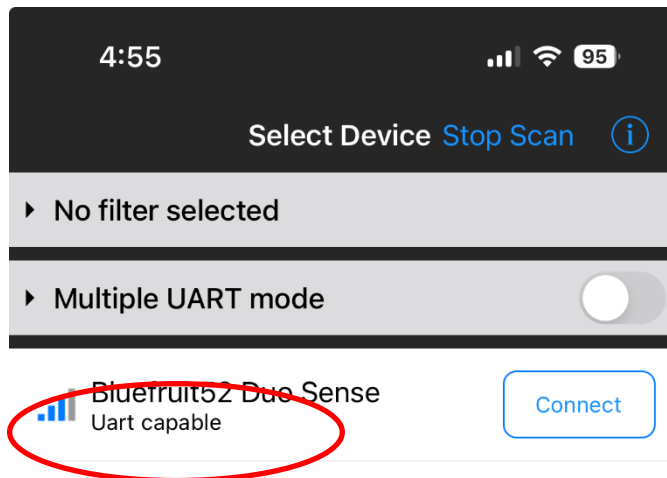
- Light + Gesture + Proximity (APDS9660).
<https://www.adafruit.com/product/3595>
- Temp + Pressure (BMP280).
<https://www.adafruit.com/product/2651>
- Magnetometer (LIS3MDL).
<https://www.adafruit.com/product/4479>
- 6DOF Gyro+Accelerometer (LSM6DS33).
<https://www.adafruit.com/product/4480>
- Humidity (SHT31).
<https://www.adafruit.com/product/2857>
- PDM MEMS Microphone.
<https://learn.adafruit.com/adafruit-pdm-microphone-breakout?view=all>
- Note that PDM microphone wasn't tested extensively by the instructors so use with care.

BLE lingo

- **BLE (Bluetooth Low Energy)** is a low power wireless protocol under Bluetooth Special Interest Group. Note that this is a different protocol from the Bluetooth Classic Standard.
- BLE device has two roles: **Central and Peripheral**. Central is similar to a Master. It initiates connections to the BLE peripherals.
- **Transactions** between a Central and Peripheral are as follows:
 - BLE **peripheral** transmits **advertisement** packets containing its name and capability while BLE **Central** perform a **scan** to detect adverts from BLE peripherals in its vicinity.
 - BLE **central** transmit **connection request** to BLE peripherals it wanted to connect to.
 - BLE **peripheral response to the connection request**, further handshaking between the Central and Peripheral results in a connection being established between them.
 - Note that this is a very simplified description omitting a number of details.

Connection to BLE Central

- You can use the [Bluefruit Connect App](#) to connect to the Adafruit Feather Sense Board to extract sensor or other data information.
- <https://learn.adafruit.com/adafruit-feather-sense/software-resources>
- Any host app that is BLE capable and [operate as a BLE Central](#) can also be used.
- The sample Arduino sketch utilise a custom [BLE UART profile](#), i.e. data exchange uses UART protocol.



ARDUINO CODE WALK THROUGH

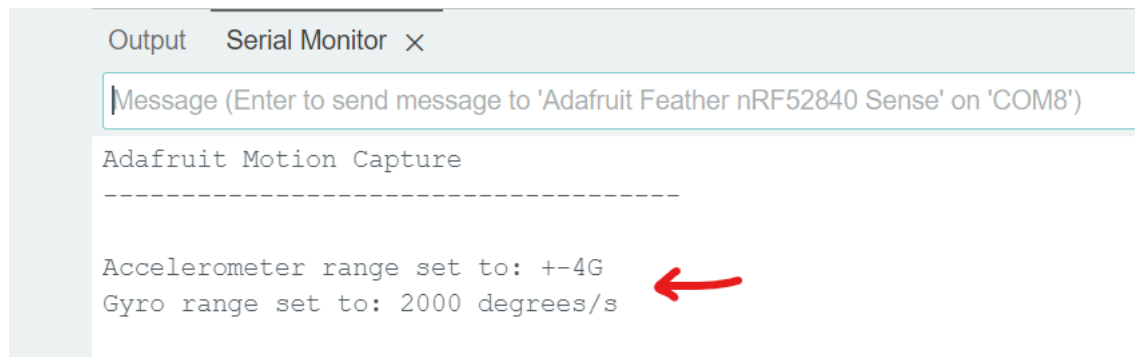
CASE STUDY#2: GESTURE CLASSIFICATION

Gesture Classification

- Similar to Gesture Classification Arduino Sketch for nano33.
- Difference is in the library used. Adafruit sensor lib are used, with different function calls.
- Difference in the output of gyro and accelerometer function calls.
- Adafruit gyro read output radian/sec instead of degree/sec for nano33.
- Adafruit accelerometer read output m/sec^2 instead of G for nano33.
- The Adafruit sketch converts radian to degree and m/sec^2 to G, so that the python code developed previously for model training (for nano33) can be used.
- The default range for accelerometer and gyro on the Adafruit board is same as that used for nano33.
 - Accelerometer: +/-4G
 - Gyro: +/-2000dps

Adafruit IMU Classifier Sketch

- Initial compilation will take some time.
- There may be some warnings regarding `memcpy` and `multiple same copies of one or more lib detected`. Ignore the `memcpy` and check if the lib(s) used is(are) from the directory you have installed the lib in.
- After uploading the file, check to see if `serial monitor` reports the `default accelerometer and gyro range`. If no printout on the Serial monitor, press the reset button. Repeat this procedure till you see the default value printout on the Serial Monitor.



The screenshot shows the Arduino IDE Serial Monitor window. The title bar says "Output Serial Monitor x". Below the title bar is a text input field with the placeholder text "Message (Enter to send message to 'Adafruit Feather nRF52840 Sense' on 'COM8')". Below the input field is the output area, which contains the text "Adafruit Motion Capture" followed by a dashed line. Below the dashed line, the output area shows the text "Accelerometer range set to: +-4G" and "Gyro range set to: 2000 degrees/s". A red arrow points to the "Gyro range set to: 2000 degrees/s" line.

```
Output Serial Monitor x
Message (Enter to send message to 'Adafruit Feather nRF52840 Sense' on 'COM8')
Adafruit Motion Capture
-----
Accelerometer range set to: +-4G
Gyro range set to: 2000 degrees/s
```


setup() and loop()

```
void setup()
{
    uint32_t i;

    Serial.begin(115200);
    while ( !Serial ) delay(10);    // for nrf52840 with native usb

    Serial.println("Adafruit Motion Capture");
    Serial.println("-----\n");

    //BLE initialisation
    //ble_init();

    //Initialise Adafruit Sensors
    init_motion();

    //Tensorflow Init
    tensorflow_init();

    Serial.println("setup done");
}

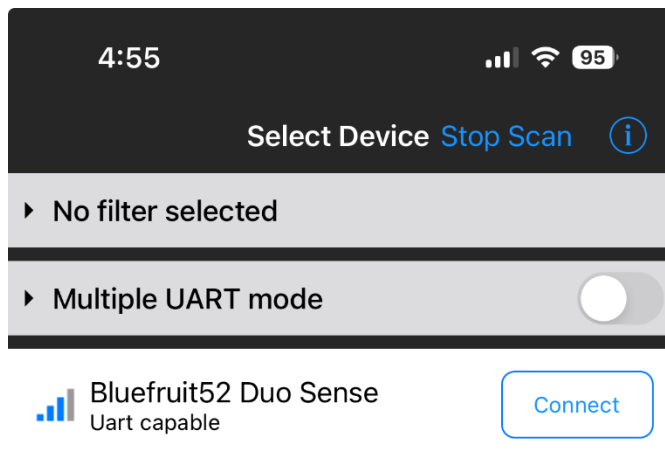
void loop()
{
    //loop_motion(); //read sensor data
    //sendBLEData(); //transmit sensor data via BLE
    loop_tensorflow(); //Model Inference Ops
}
```

- `ble_init()`: initialise for BLE operation.
- `init_motion()`: initialise all sensors on board.
- `tensorflow_init()`: initialise for Tensorflow operations.
- `loop_motion()`: read all sensor data.
- `sendBLEData()`: transmit sensor data via BLE.
- `loop_TensorFlow()`: Run tensorflowlite inference operations.
- Comment off the respective function(s) if you don't need them.

Custom BLE device name

- If you are using the BLE, you may want to give a **unique name** to your board for easy identification when you connect via the mobile phone.

```
void ble_init(void)
{
    // Initialize Bluefruit with max concurrent connections as Peripheral = 1, Central = 1
    // SRAM usage required by SoftDevice will increase with number of connections
    Bluefruit.begin(1, 1);
    Bluefruit.setTxPower(4);    // Check bluefruit.h for supported values
    Bluefruit.setName("Bluefruit52 Duo Sense"); ←
```



Assembled Prototype



- Lipo battery
- Spacers and base board for protection
- Allows more mobile applications if BLE comms is used between host and Adafruit board

ARDUINO CODE WALK THROUGH (DONE IN F2F SESSION)